

# SKILLOMATE CHANGE REPORT

## Video quality fix and certificate plan

Previous vs new implementation - 19 September 2026

### OUTCOME

The course player now prefers 1080p, uses the nearest real rendition when 1080p is unavailable, and makes quality controls reflect the resolutions actually present in the HLS manifest. A complete implementation plan is included for verifiable Skillomate completion certificates.

### WHY QUALITY DID NOT CHANGE

The interface enabled 120p through 1080p whether those renditions existed or not. It updated only HLS `currentLevel`, so the highlighted option could disagree with the stream being loaded. A source with only one rendition also cannot be switched to another resolution.

### VIDEO PLAYER CHANGES

- Default preference changed from 720p to 1080p.
- The nearest real rendition is selected when 1080p is absent.
- Missing resolutions are disabled instead of pretending they are available.
- Manual selection updates `loadLevel`, `nextLevel`, and `currentLevel` together.
- The player confirms the resolution applied.

### CERTIFICATE PLAN

The plan defines server-verified completion, idempotent issuance, unique certificate IDs, versioned PDF generation, signed downloads, QR/public verification, learner and admin interfaces, revocation/reissue audit history, mobile QA, and phased rollout.

### FILES CHANGED

edunex-f/src/components/media/CourseMediaPlayer.jsx  
edunex-f/tests/player/AwsLessonPlayer.test.jsx  
docs/CERTIFICATE-IMPLEMENTATION-PLAN.md

### VERIFICATION

Player suite: 23 files passed, 141 tests passed.  
Production frontend build: passed.  
Source formatting check: passed.

### LIMITATION

The player cannot create 1080p if the HLS master playlist does not contain a 1080p rendition. Native HLS on browsers where HLS.js is unsupported remains browser-managed.